

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	SWTID-2026-7674
Project Title	AI FAQ ASSISTANT
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to build an intelligent AI FAQ Assistant that automatically understands user queries in natural language and provides accurate, instant answers from the knowledge base using Google Gemini AI.
Scope	The project scope includes building a full-stack web application with a chat interface, backend API to manage FAQs, integration with LLM for smart answering, and an admin panel to add/edit FAQs. Scope covers data collection, API development, and cloud deployment.
Problem Statement	
Description	Businesses face high volume of repetitive customer queries, slow response times, and high manpower cost for support. Existing rule-based chatbots fail to understand natural language and give irrelevant answers, affecting customer experience.
Impact	Solving this will result in 80% reduction in support tickets, 24/7 instant customer support, reduced operational costs, and improved customer satisfaction and engagement.
Proposed Solution	
Approach	Employing Generative AI (Google Gemini 1.5 Flash) with a MERN stack. User query is sent to backend -> Backend searches MongoDB FAQs + sends context to Gemini API -> Gemini generates a human-like, accurate answer -> Displays in React chat UI. If answer not found, it gracefully handles it. Key Features AI-powered natural language understanding Instant and accurate FAQ answering Admin dashboard to Manage FAQs (Add/Edit/Delete) Secure authentication using JWT Fallback mechanism for unknown questions Responsive chat UI like ChatGPT

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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel i3 / i5, 4 Cores - Normal laptop pothum
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data	256 GB SSD
Software		
Frameworks	Web frameworks	React.js (Frontend), Node.js + Express.js (Backend)
Libraries	Additional libraries	Mongoose, JWT, bcrypt, axios, dotenv
Development Environment	IDE	VS Code, Postman, MongoDB Compass
Data and API		
Data	Source, size, format	Custom FAQ Dataset (100+ FAQs), JSON format in MongoDB
API	Third-party API	Google Gemini API (gemini-1.5-flash)